

Lecture 9

Linear regression

COURSE MATH 637: Mathematical Techniques in Data Science

WEBSITE <https://jacobcalvert.com/teaching/math637fall2026/>

INSTRUCTOR Jacob Calvert (calvert@udel.edu)

OFFICE HOURS MW 2:45pm–3:45pm, Ewing Hall 509

Announcements & Reminders

- Homework 2 is due one week from today, before class

Plan for today

- | | |
|------------------------------------|--------|
| • Regression problem | 8 min |
| • Fitting a line | 8 min |
| • Normal equation via calculus | 20 min |
| • Geometric viewpoint | 11 min |
| • Application: disease progression | 9 min |
| • Gradient descent | 10 min |

Recall: Two **unsupervised** learning problems

Problem (Dimension reduction)

Given $x_1, \dots, x_n \in \mathbb{R}^p$ and $k < p$, find an encoder $f : \mathbb{R}^p \rightarrow \mathbb{R}^k$ and a decoder $g : \mathbb{R}^k \rightarrow \mathbb{R}^p$ minimizing

$$\sum_{i=1}^n \|x_i - g(f(x_i))\|^2.$$

Problem (Density estimation)

Given $x_1, \dots, x_n \in \mathbb{R}^p$ and a family of densities $\{f_\theta : \theta \in \Theta\}$, find parameter vector $\hat{\theta} \in \Theta$ maximizing

$$\ell(\theta) = \sum_{i=1}^n \log f_\theta(x_i).$$

Each example now includes a “response”

An example is a **pair** (x_i, y_i) : the measurements $x_i \in \mathbb{R}^p$, and a number $y_i \in \mathbb{R}$ attached to them.

- x_i = patient measurements, y_i = disease progression a year later
- x_i = characteristics of a house, y_i = sale price
- x_i = settings of an instrument, y_i = quantity it is measuring

Note: x_i are the **predictors** (or features, or covariates) and y_i is the **response** (or target). This is the **supervised learning** setting.

Definition: Regression problem

Problem (Least squares regression)

Given examples $(x_1, y_1), \dots, (x_n, y_n) \in \mathbb{R}^p \times \mathbb{R}$ and a set $\mathcal{F}$ of functions $f : \mathbb{R}^p \rightarrow \mathbb{R}$, find $f \in \mathcal{F}$ minimizing the **mean squared error**

$$\text{MSE}(f) = \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i))^2.$$

Note: Just like with dimension reduction, choosing different classes $\mathcal{F}$ of functions leads to different regression problems.

Definition: Linear regression problem

Problem (Linear least squares regression)

Given examples $(x_1, y_1), \dots, (x_n, y_n) \in \mathbb{R}^p \times \mathbb{R}$, find $\theta \in \mathbb{R}^p$ minimizing

$$\text{MSE}(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - \theta^\top x_i)^2.$$

Note: Finding the best linear function means finding the p entries of θ .

Important: “Linear” refers to θ , not x !

- Suppose each example is a single number $t_i \in \mathbb{R}$
- We can build the feature vector $x_i = [1, t_i, t_i^2]^\top$
- Linear regression then fits a model that is nonlinear in t_i :

$$\theta^\top x_i = \theta_1 + \theta_2 t_i + \theta_3 t_i^2.$$

Important: “Linear” refers to θ , not x !

- Suppose each example is a single number $t_i \in \mathbb{R}$
- We can build the feature vector $x_i = [1, t_i, t_i^2]^\top$
- Linear regression then fits a model that is nonlinear in t_i :

$$\theta^\top x_i = \theta_1 + \theta_2 t_i + \theta_3 t_i^2.$$

Example (Predicting stroke risk)

Suppose you want to predict stroke risk in patients using age (a_i), blood pressure (b_i), and cholesterol (c_i). Because having high blood pressure and cholesterol together is disproportionately worse than having either on their own, you could do: $x_i = [1, a_i, b_i, c_i, (b_i c_i)^2]^\top$, so:

$$\text{risk}_i \approx \theta^\top x_i = \theta_1 + \theta_2 a_i + \theta_3 b_i + \theta_4 c_i + \theta_5 (b_i c_i)^2.$$

Convention: The design matrix

$$X = \begin{bmatrix} 1 & x_{12} & \cdots & x_{1,p} \\ 1 & x_{22} & \cdots & x_{2,p} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n2} & \cdots & x_{n,p} \end{bmatrix} = [\mathbf{1} \quad X_2 \quad \cdots \quad X_p], \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

- one **row** per example, one **column** per predictor
- the **first column is all ones**, so θ_1 is the intercept
- X is $n \times p$, and p counts the column of ones

In matrix form,

$$\text{MSE}(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - \theta^\top x_i)^2 = \frac{1}{n} \|y - X\theta\|^2.$$

Plan for today

- Regression problem 8 min
- **Fitting a line** 8 min
- Normal equation via calculus 20 min
- Geometric viewpoint 11 min
- Application: disease progression 9 min
- Gradient descent 10 min

Example: Fit a line to four points

x_i	-2	-1	1	2
y_i	1	0	2	5

Example

Find $\theta_1, \theta_2 \in \mathbb{R}$ minimizing

$$\text{MSE}(\theta) = \frac{1}{4} \sum_{i=1}^4 (y_i - \theta_1 - \theta_2 x_i)^2.$$

Here $p = 2$: the design matrix is $X = \begin{bmatrix} 1 & x \end{bmatrix}$, with a column of ones and a column holding $x_1, \dots, x_4$.

Example: Fit a line to four points

- Set both partial derivatives to zero, writing ∂_{θ_j} for $\partial/\partial\theta_j$:

$$\partial_{\theta_1} \text{MSE} = -\frac{2}{n} \sum_i (y_i - \theta_1 - \theta_2 x_i) = 0,$$

$$\partial_{\theta_2} \text{MSE} = -\frac{2}{n} \sum_i x_i (y_i - \theta_1 - \theta_2 x_i) = 0.$$

Example: Fit a line to four points

- Set both partial derivatives to zero, writing ∂_{θ_j} for $\partial/\partial\theta_j$:

$$\partial_{\theta_1} \text{MSE} = -\frac{2}{n} \sum_i (y_i - \theta_1 - \theta_2 x_i) = 0,$$

$$\partial_{\theta_2} \text{MSE} = -\frac{2}{n} \sum_i x_i (y_i - \theta_1 - \theta_2 x_i) = 0.$$

- The factor $-2/n$ divides out, and the two equations say

$$\sum_i y_i = n\theta_1 + \theta_2 \sum_i x_i, \quad \sum_i x_i y_i = \theta_1 \sum_i x_i + \theta_2 \sum_i x_i^2.$$

Example: Fit a line to four points

- Set both partial derivatives to zero, writing ∂_{θ_j} for $\partial/\partial\theta_j$:

$$\partial_{\theta_1} \text{MSE} = -\frac{2}{n} \sum_i (y_i - \theta_1 - \theta_2 x_i) = 0,$$

$$\partial_{\theta_2} \text{MSE} = -\frac{2}{n} \sum_i x_i (y_i - \theta_1 - \theta_2 x_i) = 0.$$

- The factor $-2/n$ divides out, and the two equations say

$$\sum_i y_i = n\theta_1 + \theta_2 \sum_i x_i, \quad \sum_i x_i y_i = \theta_1 \sum_i x_i + \theta_2 \sum_i x_i^2.$$

- Solving the pair,

$$\hat{\theta}_2 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \quad \hat{\theta}_1 = \bar{y} - \hat{\theta}_2 \bar{x}.$$

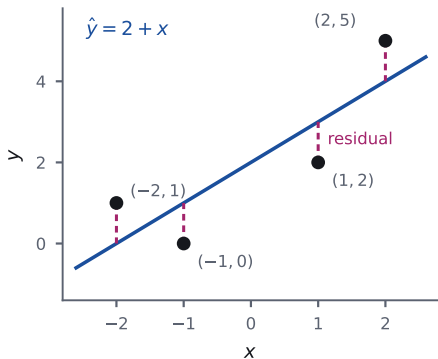
Example: Fit a line to four points

Here $\bar{x} = 0$ and $\bar{y} = 2$, so

$$\hat{\theta}_2 = \frac{\sum_i x_i y_i}{\sum_i x_i^2} = \frac{10}{10} = 1, \quad \hat{\theta}_1 = \bar{y} = 2.$$

The predicted values $\hat{y}$, residual $y - \hat{y}$, and resulting MSE are:

- $\hat{y} = (0, 1, 3, 4)$
- $r = y - \hat{y} = (1, -1, -1, 1)$
- $\text{MSE}(\hat{\theta}) = \frac{1}{4} \|r\|^2 = 1$



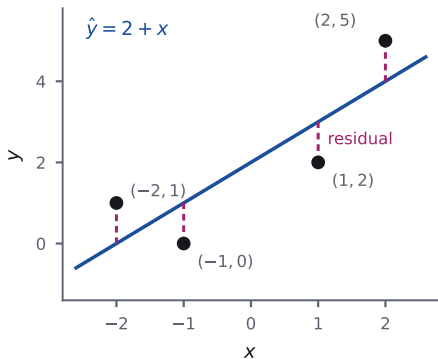
Example: Fit a line to four points

Here $\bar{x} = 0$ and $\bar{y} = 2$, so

$$\hat{\theta}_2 = \frac{\sum_i x_i y_i}{\sum_i x_i^2} = \frac{10}{10} = 1, \quad \hat{\theta}_1 = \bar{y} = 2.$$

The predicted values $\hat{y}$, residual $y - \hat{y}$, and resulting MSE are:

- $\hat{y} = (0, 1, 3, 4)$
- $r = y - \hat{y} = (1, -1, -1, 1)$
- $\text{MSE}(\hat{\theta}) = \frac{1}{4} \|r\|^2 = 1$



Idea: For $p > 2$, instead of taking the partial derivatives of MSE w.r.t. each θ_j separately, we want to find $\nabla \text{MSE}(\theta)$.

Plan for today

- Regression problem 8 min
- Fitting a line 8 min
- Normal equation via calculus 20 min
- Geometric viewpoint 11 min
- Application: disease progression 9 min
- Gradient descent 10 min

Recall: The gradient

Definition (Derivative with respect to a vector)

For $f : \mathbb{R}^p \rightarrow \mathbb{R}$, the **gradient** of f at θ is the vector of partial derivatives:

$$\nabla f(\theta) = \begin{bmatrix} \partial_{\theta_1} f(\theta) \\ \vdots \\ \partial_{\theta_p} f(\theta) \end{bmatrix} \in \mathbb{R}^p,$$

where $\partial_{\theta_j} f$ is the ordinary derivative in θ_j with the other entries fixed.

Aside: Some derivatives w.r.t. a vector

- If $f(\theta) = \theta^\top a$ for $a \in \mathbb{R}^p$ fixed, then $\nabla f(\theta) = a$ because

$$\theta^\top a = \sum_k \theta_k a_k, \quad \text{so} \quad \partial_{\theta_j}(\theta^\top a) = \sum_k (\partial_{\theta_j} \theta_k) a_k = \sum_k \delta_{jk} a_k = a_j.$$

Aside: Some derivatives w.r.t. a vector

- If $f(\theta) = \theta^\top a$ for $a \in \mathbb{R}^p$ fixed, then $\nabla f(\theta) = a$ because

$$\theta^\top a = \sum_k \theta_k a_k, \quad \text{so} \quad \partial_{\theta_j}(\theta^\top a) = \sum_k (\partial_{\theta_j} \theta_k) a_k = \sum_k \delta_{jk} a_k = a_j.$$

- If $f(\theta) = \theta^\top A \theta$ for **symmetric** $A \in \mathbb{R}^{p \times p}$, then $\nabla f(\theta) = 2A\theta$ because

$$\theta^\top A \theta = \sum_{k,l} \theta_k A_{kl} \theta_l, \quad \text{so}$$

$$\partial_{\theta_j}(\theta^\top A \theta) = \sum_{k,l} \delta_{jk} A_{kl} \theta_l + \sum_{k,l} \theta_k A_{kl} \delta_{jl} = \sum_l A_{jl} \theta_l + \sum_k \underbrace{\theta_k A_{kj}}_{=A_{jk} \theta_k} = 2(A\theta)_j.$$

Aside: Some derivatives w.r.t. a vector

- If $a \in \mathbb{R}^p$ fixed, then $\nabla(\theta^\top a) = a$ because

$$\theta^\top a = \sum_k \theta_k a_k, \quad \text{so} \quad \partial_{\theta_j}(\theta^\top a) = \sum_k (\partial_{\theta_j} \theta_k) a_k = \sum_k \delta_{jk} a_k = a_j.$$

- If $A \in \mathbb{R}^{p \times p}$ **symmetric**, then $\nabla(\theta^\top A \theta) = 2A\theta$ because

$$\theta^\top A \theta = \sum_{k,l} \theta_k A_{kl} \theta_l, \quad \text{so}$$

$$\partial_{\theta_j}(\theta^\top A \theta) = \sum_{k,l} \delta_{jk} A_{kl} \theta_l + \sum_{k,l} \theta_k A_{kl} \delta_{jl} = \sum_l A_{jl} \theta_l + \sum_k \underbrace{\theta_k A_{kj}}_{=A_{jk} \theta_k} = 2(A\theta)_j.$$

Idea: Generalizations of $(\theta_1 a_1)' = a_1$ and $(a_1 \theta_1^2)' = 2a_1 \theta_1$ to $p > 1$.

Calculation: The gradient of the MSE

- Expand the squared norm, using $\|v\|^2 = v^\top v$ and $(X\theta)^\top y = \theta^\top X^\top y$:

$$\text{MSE}(\theta) = \frac{1}{n} \|y - X\theta\|^2 = \frac{1}{n} \left(y^\top y - 2\theta^\top X^\top y + \theta^\top X^\top X \theta \right).$$

Calculation: The gradient of the MSE

- Expand the squared norm, using $\|v\|^2 = v^\top v$ and $(X\theta)^\top y = \theta^\top X^\top y$:

$$\text{MSE}(\theta) = \frac{1}{n} \|y - X\theta\|^2 = \frac{1}{n} \left(y^\top y - 2\theta^\top X^\top y + \theta^\top X^\top X \theta \right).$$

- Apply the rules from the previous slide with $a = X^\top y$ and $A = X^\top X$:

$$\nabla \text{MSE}(\theta) = \frac{1}{n} (-2 X^\top y + 2 X^\top X \theta) = \frac{2}{n} X^\top (X\theta - y).$$

Calculation: The gradient of the MSE

- Expand the squared norm, using $\|v\|^2 = v^\top v$ and $(X\theta)^\top y = \theta^\top X^\top y$:

$$\text{MSE}(\theta) = \frac{1}{n} \|y - X\theta\|^2 = \frac{1}{n} \left(y^\top y - 2\theta^\top X^\top y + \theta^\top X^\top X \theta \right).$$

- Apply the rules from the previous slide with $a = X^\top y$ and $A = X^\top X$:

$$\nabla \text{MSE}(\theta) = \frac{1}{n} (-2 X^\top y + 2 X^\top X \theta) = \frac{2}{n} X^\top (X\theta - y).$$

- Solving $\nabla \text{MSE}(\hat{\theta}) = 0$ gives the “normal equation”

$$X^\top X \hat{\theta} = X^\top y.$$

Calculation: The gradient of the MSE

- Expand the squared norm, using $\|v\|^2 = v^\top v$ and $(X\theta)^\top y = \theta^\top X^\top y$:

$$\text{MSE}(\theta) = \frac{1}{n} \|y - X\theta\|^2 = \frac{1}{n} \left(y^\top y - 2\theta^\top X^\top y + \theta^\top X^\top X \theta \right).$$

- Apply the rules from the previous slide with $a = X^\top y$ and $A = X^\top X$:

$$\nabla \text{MSE}(\theta) = \frac{1}{n} (-2 X^\top y + 2 X^\top X \theta) = \frac{2}{n} X^\top (X\theta - y).$$

- Solving $\nabla \text{MSE}(\hat{\theta}) = 0$ gives the “normal equation”

$$X^\top X \hat{\theta} = X^\top y.$$

Note: The normal equation may have more than one solution!

Normal equation identifies global minimizer of MSE

Theorem

$\hat{\theta}$ minimizes $\text{MSE}(\theta)$ over $\theta \in \mathbb{R}^p$ if and only if $X^T X \hat{\theta} = X^T y$.

Normal equation identifies global minimizer of MSE

Theorem

$\hat{\theta}$ minimizes $\text{MSE}(\theta)$ over $\theta \in \mathbb{R}^p$ if and only if $X^\top X \hat{\theta} = X^\top y$.

Proof. Let $\hat{\theta}$ satisfy $X^\top(y - X\hat{\theta}) = 0$. For any $\theta \in \mathbb{R}^p$, we have

$$y - X\theta = (y - X\hat{\theta}) + X(\hat{\theta} - \theta).$$



Normal equation identifies global minimizer of MSE

Theorem

$\hat{\theta}$ minimizes $\text{MSE}(\theta)$ over $\theta \in \mathbb{R}^p$ if and only if $X^T X \hat{\theta} = X^T y$.

Proof. Let $\hat{\theta}$ satisfy $X^T(y - X\hat{\theta}) = 0$. For any $\theta \in \mathbb{R}^p$, we have

$$y - X\theta = (y - X\hat{\theta}) + X(\hat{\theta} - \theta).$$

By expanding the square, we find that

$$\|y - X\theta\|^2 = \|y - X\hat{\theta}\|^2 + 2(\hat{\theta} - \theta)^T X^T (y - X\hat{\theta}) + \|X(\hat{\theta} - \theta)\|^2.$$



Normal equation identifies global minimizer of MSE

Theorem

$\hat{\theta}$ minimizes $\text{MSE}(\theta)$ over $\theta \in \mathbb{R}^p$ if and only if $X^T X \hat{\theta} = X^T y$.

Proof. Let $\hat{\theta}$ satisfy $X^T(y - X\hat{\theta}) = 0$. For any $\theta \in \mathbb{R}^p$, we have

$$y - X\theta = (y - X\hat{\theta}) + X(\hat{\theta} - \theta).$$

By expanding the square, we find that

$$\|y - X\theta\|^2 = \|y - X\hat{\theta}\|^2 + 2(\hat{\theta} - \theta)^T X^T (y - X\hat{\theta}) + \|X(\hat{\theta} - \theta)\|^2.$$

The middle term is zero, so $\text{MSE}(\theta) \geq \text{MSE}(\hat{\theta})$ for every θ . □

What $\hat{\theta}$ and $\hat{y}$ become

If $X^T X$ is invertible, $X^T X \hat{\theta} = X^T y$ has exactly one solution:

$$\hat{\theta} = (X^T X)^{-1} X^T y, \quad \hat{y} = X \hat{\theta} = X (X^T X)^{-1} X^T y.$$

What $\hat{\theta}$ and $\hat{y}$ become

If $X^T X$ is invertible, $X^T X \hat{\theta} = X^T y$ has exactly one solution:

$$\hat{\theta} = (X^T X)^{-1} X^T y, \quad \hat{y} = X \hat{\theta} = X (X^T X)^{-1} X^T y.$$

Example (The four points)

$X^T X = \text{diag}(4, 10)$ and $X^T y = (8, 10)^T$, so

$$\hat{\theta} = \left(\frac{8}{4}, \frac{10}{10}\right) = (2, 1), \quad \hat{y} = (0, 1, 3, 4).$$

$X^T X$ is diagonal because $\mathbf{1}^T x = \sum_i x_i = 0$.

Note: Whether $X^T X$ is invertible is a topic for next lecture.

How good is $\hat{\theta}$? Compare MSE to the constant model

Compare the fit against the simplest predictor there is: a **constant** c , using no features at all.

This is least squares with $X = \mathbf{1}$, a single column, and $\hat{c} = \bar{y}$, so

$$\text{MSE}_{\text{const}} = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2.$$

How good is $\hat{\theta}$? Compare MSE to the constant model

Compare the fit against the simplest predictor there is: a **constant** c , using no features at all.

This is least squares with $X = \mathbf{1}$, a single column, and $\hat{c} = \bar{y}$, so

$$\text{MSE}_{\text{const}} = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2.$$

Example

On the four points, $\bar{y} = 2$, so $\text{MSE}_{\text{const}} = \frac{1}{4}(1 + 4 + 0 + 9) = 3.5$, against $\text{MSE}(\hat{\theta}) = 1$ for the fitted line.

How good is $\hat{\theta}$? R^2

Definition (Coefficient of determination)

$$R^2 = 1 - \frac{\text{MSE}(\hat{\theta})}{\text{MSE}_{\text{const}}},$$

the fraction of the constant model's error that the features removed.

How good is $\hat{\theta}$? R^2

Definition (Coefficient of determination)

$$R^2 = 1 - \frac{\text{MSE}(\hat{\theta})}{\text{MSE}_{\text{const}}},$$

the fraction of the constant model's error that the features removed.

Example

On the four points, $R^2 = 1 - 1/3.5 = \frac{5}{7} \approx 0.71$.

How good is $\hat{\theta}$? R^2

Definition (Coefficient of determination)

$$R^2 = 1 - \frac{\text{MSE}(\hat{\theta})}{\text{MSE}_{\text{const}}} = 1 - \frac{\text{RSS}}{\text{TSS}},$$

the fraction of the constant model's error that the features removed.

You may be more familiar with R^2 expressed in terms of:

- Residual sum of squares (RSS) = $\sum_i (y_i - \hat{y}_i)^2$
- Total sum of squares (TSS) = $\sum_i (y_i - \bar{y})^2$

Plan for today

- Regression problem 8 min
- Fitting a line 8 min
- Normal equation via calculus 20 min
- **Geometric viewpoint** 11 min
- Application: disease progression 9 min
- Gradient descent 10 min

A geometric view of least squares

Note that $X\theta$ is a linear combination of the columns X_i , weighted by the θ_i :

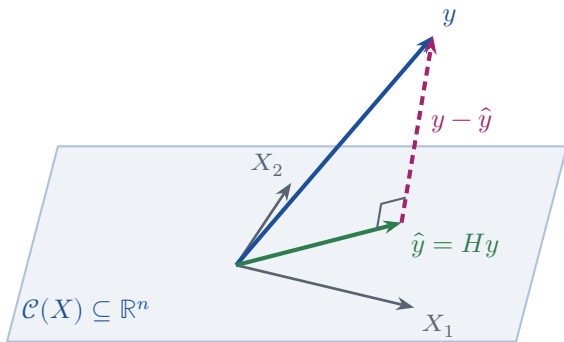
$$X\theta = \sum_{i=1}^p \theta_i X_i.$$

So $X\theta$ ranges over the **column space** of X

$$\mathcal{C}(X) = \text{span}\{X_1, \dots, X_p\} \subseteq \mathbb{R}^n, \quad \dim \mathcal{C}(X) = \text{rank}(X) \leq p.$$

Key idea: An exact solution of $y = X\theta$ exists when $y \in \mathcal{C}(X)$. Otherwise, we should project y onto $\mathcal{C}(X)$.

Least squares is a projection



Minimizing $\|y - X\theta\|$ over $\theta \in \mathbb{R}^p$ is the same as minimizing $\|y - v\|$ over $v \in \mathcal{C}(X)$.

Key idea: The closest point of $\mathcal{C}(X)$ to y is the orthogonal projection $\hat{y}$ onto it. So we can get $\hat{\theta}$ a second way: by writing the matrix H that projects onto $\mathcal{C}(X)$.

Recall: Projection matrix

If $v_1, \dots, v_k$ are **orthonormal** and $V = [v_1 \ \dots \ v_k]$, then the projection $\hat{y}$ of y onto $\text{span}\{v_1, \dots, v_k\}$ is

$$\hat{y} = VV^T y = \sum_{j=1}^k v_j (v_j^T y).$$

Recall: Projection matrix

If $v_1, \dots, v_k$ are **orthonormal** and $V = [v_1 \ \dots \ v_k]$, then the projection $\hat{y}$ of y onto $\text{span}\{v_1, \dots, v_k\}$ is

$$\hat{y} = VV^T y = \sum_{j=1}^k v_j (v_j^T y).$$

Key idea: The columns of X are neither orthogonal nor of unit length, so $XX^T y$ **does not** project y onto $\mathcal{C}(X)$! **Assuming $X^T X$ is invertible**, we can fix this by putting $(X^T X)^{-1}$ in the middle:

$$H = X(X^T X)^{-1} X^T.$$

H is sometimes called the “hat” matrix, because $\hat{y} = Hy$.

Two derivations, one system

From calculus

Set $\nabla \text{MSE}(\theta) = 0$. This method also works for other error measures and other classes of regression functions.

From geometry

Project y onto $\mathcal{C}(X)$. Relies on fact that MSE based on $\|\cdot\|$. Other ways to write H when $X^T X$ not invertible.

Key idea: Both approaches give $X^T X \hat{\theta} = X^T y$.

Watch out: $\hat{\theta} = (X^T X)^{-1} X^T y$ is a formula, not an algorithm. E.g., the inverse may not exist or $X^T X$ may be too large to work with. Can use gradient descent to find $\hat{\theta}$ instead.

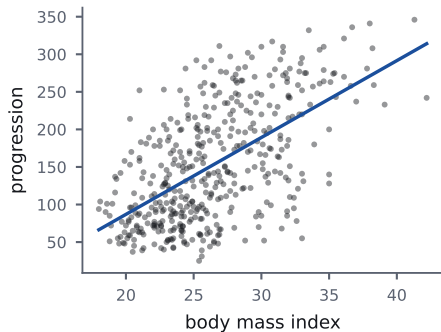
Plan for today

- Regression problem 8 min
- Fitting a line 8 min
- Normal equation via calculus 20 min
- Geometric viewpoint 11 min
- **Application: disease progression** 9 min
- Gradient descent 10 min

Application: Disease progression

From $n = 442$ diabetes patients:

- each x_i has 10 measurements:
 1. age
 2. blood pressure
 3. body mass index
 4. sex
 - 5–10. six blood serum measurements (“s1” through “s6”)
- the response y_i : a measure of disease progression one year later



Efron, Hastie, Johnstone and Tibshirani (2004)

Application: Disease progression

```
from sklearn.datasets import load_diabetes  
  
d = load_diabetes(scaled=False)  
n = len(d.target)  
X = np.column_stack([np.ones(n), d.data])  
theta, *_ = np.linalg.lstsq(X, d.target, rcond=None)
```

- `np.ones(n)` is the intercept column, so X is 442×11

Application: Disease progression

```
from sklearn.datasets import load_diabetes

d = load_diabetes(scaled=False)
n = len(d.target)
X = np.column_stack([np.ones(n), d.data])
theta, *_ = np.linalg.lstsq(X, d.target, rcond=None)
```

- `np.ones(n)` is the intercept column, so X is 442×11
- `lstsq` solves the least squares problem directly, without forming $X^T X$

Application: Disease progression

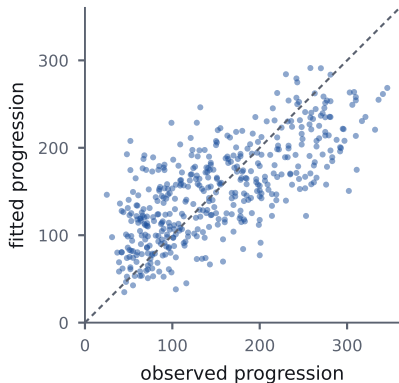
intercept	-334.57	s1	-1.09
age	-0.04	s2	0.75
sex	-22.86	s3	0.37
bmi	5.60	s4	6.53
bp	1.12	s5	68.48
		s6	0.28

Note: Each coefficient is in units of progression per unit of its own predictor, so the sizes reflect the units of the columns as much as anything else: `bmi` runs from 18 to 42, and `s5` from 3.26 to 6.11.

Application: Disease progression

After fitting $\hat{\theta}$:

- $\text{MSE}(\hat{\theta}) = 2860$, so typical error in predicted progression is $\sqrt{2860}$
- Compare this to $\text{MSE}_{\text{const}} = 5930$
- $R^2 = 1 - 2860/5930 = 0.518$



Application: Disease progression

Serum measurement `s1` has correlation $+0.21$ with progression, and coefficient -1.09 in the fit.

- `s1` and `s2` have correlation 0.90

Watch out: Interpreting the coefficients can be tricky because the features can depend on one another. In particular, this is true of the higher-order features we discussed earlier!

Application: Disease progression

Serum measurement `s1` has correlation $+0.21$ with progression, and coefficient -1.09 in the fit.

- `s1` and `s2` have correlation 0.90
- the coefficient of `s1` describes changing `s1` with the other nine predictors **held fixed**

Watch out: Interpreting the coefficients can be tricky because the features can depend on one another. In particular, this is true of the higher-order features we discussed earlier!

Plan for today

- Regression problem 8 min
- Fitting a line 8 min
- Normal equation via calculus 20 min
- Geometric viewpoint 11 min
- Application: disease progression 9 min
- Gradient descent 10 min

Complexity of solving the normal equation

To solve $X^T X \hat{\theta} = X^T y$ we must form $X^T X$ and then factor it:

$$\underbrace{np^2}_{\text{form } X^T X} + \underbrace{p^3}_{\text{factor it}} \quad \text{arithmetic operations.}$$

- at $n = 10^6$ and $p = 10^4$, forming $X^T X$ alone is 10^{14} operations
- and $X^T X$ is $10^4 \times 10^4$, which has to be stored

Idea: Computing $\nabla \text{MSE}(\theta) = \frac{2}{n} X^T (X\theta - y)$ is two matrix-vector products, or $2np$ operations, which is 2×10^{10} here. We could calculate the gradient 5000 times for the price of forming $X^T X$ once.

Algorithm: Gradient descent

Definition (Gradient descent for least squares)

Choose $\theta^{(0)}$ and a step size $\eta > 0$, and repeat

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla \text{MSE}(\theta^{(t)}) = \theta^{(t)} - \frac{2\eta}{n} X^{\top} (X\theta^{(t)} - y).$$

- each step uses one product $X\theta^{(t)}$ and one product with $X^{\top}$

Algorithm: Gradient descent

Definition (Gradient descent for least squares)

Choose $\theta^{(0)}$ and a step size $\eta > 0$, and repeat

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla \text{MSE}(\theta^{(t)}) = \theta^{(t)} - \frac{2\eta}{n} X^{\top} (X\theta^{(t)} - y).$$

- each step uses one product $X\theta^{(t)}$ and one product with $X^{\top}$
- we never need to form $X^{\top}X$

Algorithm: Gradient descent

Definition (Gradient descent for least squares)

Choose $\theta^{(0)}$ and a step size $\eta > 0$, and repeat

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla \text{MSE}(\theta^{(t)}) = \theta^{(t)} - \frac{2\eta}{n} X^T (X\theta^{(t)} - y).$$

- each step uses one product $X\theta^{(t)}$ and one product with X^T
- we never need to form $X^T X$
- we get to choose η , but not all choices work

Why **gradient descent** minimizes MSE

- If $\hat{\theta}$ solves the normal equations then $\nabla \text{MSE}(\theta) = \frac{2}{n} X^T X (\theta - \hat{\theta})$, so subtracting $\hat{\theta}$ from both sides of the update gives

$$\theta^{(t+1)} - \hat{\theta} = \left(I - \frac{2\eta}{n} X^T X \right) (\theta^{(t)} - \hat{\theta}).$$

Why **gradient descent** minimizes MSE

- If $\hat{\theta}$ solves the normal equations then $\nabla \text{MSE}(\theta) = \frac{2}{n} X^T X (\theta - \hat{\theta})$, so subtracting $\hat{\theta}$ from both sides of the update gives

$$\theta^{(t+1)} - \hat{\theta} = \left(I - \frac{2\eta}{n} X^T X \right) (\theta^{(t)} - \hat{\theta}).$$

- In an orthonormal eigenbasis of $X^T X$, the component with eigenvalue λ is multiplied by $1 - 2\eta\lambda/n$ each step. It shrinks exactly when

$$\left| 1 - \frac{2\eta\lambda}{n} \right| < 1 \quad \Longleftrightarrow \quad 0 < \eta < \frac{n}{\lambda},$$

and this must hold for every λ , so $\lambda_{\max}$ decides: $0 < \eta < n/\lambda_{\max}$.

Why **gradient descent** minimizes MSE

- If $\hat{\theta}$ solves the normal equations then $\nabla \text{MSE}(\theta) = \frac{2}{n} X^T X (\theta - \hat{\theta})$, so subtracting $\hat{\theta}$ from both sides of the update gives

$$\theta^{(t+1)} - \hat{\theta} = \left(I - \frac{2\eta}{n} X^T X \right) (\theta^{(t)} - \hat{\theta}).$$

- In an orthonormal eigenbasis of $X^T X$, the component with eigenvalue λ is multiplied by $1 - 2\eta\lambda/n$ each step. It shrinks exactly when

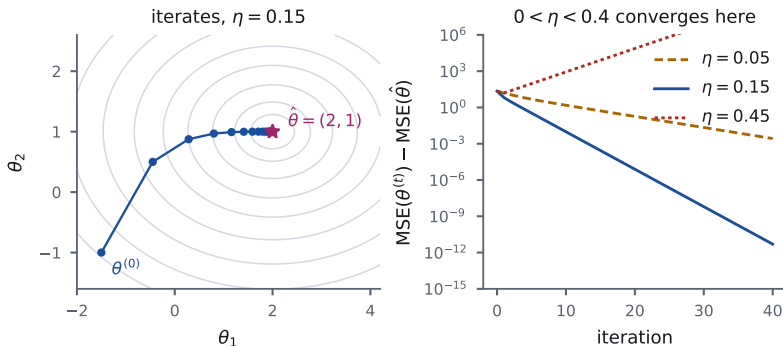
$$\left| 1 - \frac{2\eta\lambda}{n} \right| < 1 \quad \Longleftrightarrow \quad 0 < \eta < \frac{n}{\lambda},$$

and this must hold for every λ , so $\lambda_{\max}$ decides: $0 < \eta < n/\lambda_{\max}$.

- MSE is convex, so the limit is a **global** minimum

Example: The four points again

$X^T X = \text{diag}(4, 10)$ and $n = 4$, so the iterates converge when $0 < \eta < 0.4$. At $\eta = 0.15$ the factors are 0.70 and 0.25, and $\text{MSE} = 1.0098$ after ten steps against the exact 1.



What gradient descent is really for

Here we have a closed form, so descending to it is a convenience. Two things change that:

- p or n large enough that $X^T X$ cannot be formed or stored

Note: From $\theta^{(0)} = 0$ the iterates never move in a direction that leaves $X\theta$ unchanged, so the limit is the minimizer of smallest norm.

Coming up: A random handful of rows per step gives stochastic gradient descent.

What gradient descent is really for

Here we have a closed form, so descending to it is a convenience. Two things change that:

- p or n large enough that $X^T X$ cannot be formed or stored
- a model with **no closed form at all**: change $\mathcal{F}$, or change the squared error, and the normal equations have no analogue

Note: From $\theta^{(0)} = 0$ the iterates never move in a direction that leaves $X\theta$ unchanged, so the limit is the minimizer of smallest norm.

Coming up: A random handful of rows per step gives stochastic gradient descent.